MANDOWN V5

Vertical/Foreshortened Fall Detection — Full Investigation, Architecture Reference & Real-World Validation

Prepared: September 2026 | Camera(s) under test: two live RTSP CCTV feeds (referred to as Camera 1 / Camera 2) plus four recorded reference videos

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1. Executive Summary

This report documents the effort to make the MANDOWN (fall-detection) pipeline correctly recognize a person who has fallen **toward or away from the camera** — a "vertical" or "foreshortened" fall, where the body's long axis points roughly along the camera's own line of sight instead of across it. This orientation is a fundamentally hard case for any 2D-keypoint-based system: a body lying flat toward the camera and a body standing upright both project to a similar narrow, vertical shape in the 2D image, so any feature computed purely from 2D pixel angles is structurally blind to the difference.

Over this engineering round, the pipeline gained two new structural capabilities that did not exist before:

- **A proper 2D (row + column) ground-plane perspective calibration**, replacing a 1-D (row-only) fit, so expected body size is compared against the right expectation for a person's actual position in frame, not just their vertical position.
- **Monocular 2D→3D pose lifting** (via Google's MediaPipe Pose Landmarker), giving the system a real, camera-orientation-independent estimate of body tilt — something no 2D-only feature can provide by construction.

Alongside these, seven distinct real false positives and false negatives were found through live RTSP testing (not synthetic data) and fixed at the root, with two more attempted fixes found unsafe and

reverted before shipping once real regression testing caught the problem. The system's most safety-critical historical regression point — a specific real recorded fall in a validation video — was re-protected and re-verified after every change in this round.

As of this report, the vertical-fall case is **substantially improved and confirmed working for several real tested postures** (see §6), but **one further real posture variant (lying with one knee raised) has been found still unconfirmed** and remains open — see §7 for the exact numbers and §10 for the recommended next step.

2. System Architecture Overview

The pipeline is a chain of independent stages. Each stage's output feeds the next; no stage is allowed to silently fabricate a value it cannot support with real evidence — this "never fabricate, prefer honest neutral defaults" discipline appears repeatedly through the codebase and is treated as a hard design rule.

```
Video frame (RTSP / file)
      │
      ▼
┌──────────────────────────────┐
│ YOLO11-pose detector         │  → per-person bounding box + 17 COCO keypoints (x, y,
confidence)
│ + BoT-SORT tracker           │  → a stable track_id per physical person across frames
└──────────────────────────────┘
      │
      ├──────────────────────────────────────────────────────┐
      ▼                                                      ▼
┌──────────────────────────────┐                    ┌──────────────────────────────────┐
│ Base FeatureExtractor        │                    │ ExtendedFeatureExtractor
│
│ (5 features, UNCHANGED       │                    │ (torso_angle, hip_angle,
│
│  since project start)        │                    │  body_axis_angle, leg
│
│                              │                    │  extension, body_height_norm,
│
│  → feeds the pretrained      │                    │  torso_angle_3d, ...)
│
│    30-frame LSTM as-is       │                    └──────────────────────────────────┘
└──────────────────────────────┘                                   │
      │                                                            ▼
      ▼                                                 ┌──────────────────────────────────┐
┌──────────────────────────────┐                        │ FeatureMandownScorer
│
│ 5-feature LSTM               │                        │ (down_score, floor_proximity,
│
│ (models/fall_detector.py,    │                        │  kneeling_penalty, height_
```

```
│
│  best.pth checkpoint)    │                          │  collapse_cue, persistence,
│
│  → lstm_mandown_prob     │                          │  pose_quality, fall_motion)
│
└──────────────────────────┘                          └──────────────────────────────┘
      │                                                              │
      └──────────────────────────────┬───────────────────────────────┘
                                     ▼
                    ┌──────────────────────────────────┐
                    │ EvidenceFusion                   │  → fusion_score, suggested_tier
                    │ (weighted combination + a        │     (NORMAL / POSSIBLE_MANDOWN /
MANDOWN)
                    │  set of override/cap rules)      │
                    └──────────────────────────────────┘
                                     │
                                     ▼
                    ┌──────────────────────────────────┐
                    │ TrackStateMachine                │  → per-track hysteresis:
                    │ (per track_id, sticky states)    │     needs N consecutive "up"
frames
                    └──────────────────────────────────┘     to confirm, M consecutive
                                     │                         "normal" frames to recover
                                     ▼
                    ┌──────────────────────────────────┐
                    │ FallDetectionPipeline            │  → drawing, banners, event
dedup,
                    │ (src/pipeline.py)                │     multi-person screenshots,
JSON log
                    └──────────────────────────────────┘
```

Two camera-calibration subsystems run alongside this, continuously refining themselves from real observed people (never from a hard-coded assumption about the camera):

- **SceneProfileBuilder** (`src/scene_profile.py`) — learns, from many confidently-standing people observed over time, how a person's expected pixel height/width varies across the image, and where on-screen the floor actually is.
- **RollingPersonSizeStats** — a simpler, position-agnostic fallback used only until the position-aware fit above has enough samples.

3. The MANDOWN Detection Logic, In Detail

3.1 Base features (LSTM input) — untouched by design

The original 5-feature LSTM classifier (`body_angle` , `cog_height` , `vertical_velocity` , `bbox_aspect_ratio` , `keypoint_confidence`) and its trained checkpoint have been treated as a hard

constraint throughout this and prior engineering rounds: never retrained, never given a different input contract. Every fix described in this report works entirely in a deterministic geometric layer built *on top of* the LSTM, so the LSTM's ~97-98% test-set accuracy on its original task is never put at risk. The LSTM's output (`lstm_mandown_prob`) is treated as one supporting opinion among several, not a gate — this was a deliberate decision from the V3 engineering round, since an already-lying person with a cold (not-yet-filled) or simply wrong LSTM buffer must still be detectable from geometry alone.

3.2 Extended geometric features

Computed independently, once per person per frame, from the same 17 keypoints:

Feature	Meaning
`torso_angle`	Angle of the shoulder-midpoint→hip-midpoint line from vertical, 0 (vertical) to 1 (horizontal). Generalizes correctly to any 2D rotation direction.
`hip_angle`	Same idea, for the hip-midpoint→knee-midpoint line. Corrupted by knee bend, so used carefully (see §5.7).
`body_axis_angle`	Long-baseline head→ankle-midpoint axis angle. Corroborates `torso_angle` only when the torso already shows non-trivial lean (agreement-gated, never a blind override).
`leg_extension_ratio`	How straight the leg is (hip-knee-ankle path length vs. direct hip-ankle distance). Low = bent knee.
`body_height_norm`	Observed bounding-box height ÷ this camera's calibrated expected standing height at this exact image position. ~1.0 = normal, well below 1.0 = shorter than a standing person should be here.
`cog_height`	Normalized hip height in the frame (0 = top, 1 = bottom) — the basis for the self-relative "hip drop" signal (§5.3).
`torso_angle_3d` NEW this round	Angle between the real (lifted) 3D shoulder→hip vector and true vertical. See §5.8.

3.3 FeatureMandownScorer — deterministic geometric scoring

Combines the extended features into several independent 0–1 scores:

- **`down_score`** — the primary "is this posture down" signal: a blend of body shape (bounding-box aspect ratio), a torso-only orientation cue (a boundary sigmoid past ~36° from vertical, not a linear ramp to 90°), and a calibrated height-collapse cue, each contributing 40/40/20% before an agreement-gated corroboration boost.
- **`floor_proximity_score`** — head-to-knee vertical span collapsing, corroborated by the same orientation and height-collapse cues.
- **`kneeling_penalty`** — a multiplicative gate (0 = no suppression, 1 = full suppression) applied to both scores above, specifically to stop kneeling/sitting/crouching (bent-leg) postures and — after

this round's fixes — several other "this looks down but isn't actually lying" postures from registering as a fall.

- `height_collapse_cue` — orientation-independent: is the observed body shorter than this camera's calibrated expectation for a standing person at this row/column? Exposed separately because it is the one signal that does not depend on angle at all.
- `persistence_score` — fraction of the recent window reading consistently "down".
- `fall_motion_score` — a recent spike in downward velocity/angle-change, supporting evidence only (never required).
- `pose_quality_score` — mean keypoint confidence, used to discount (never zero out) confidence from an unreliable skeleton.

3.4 EvidenceFusion — combining LSTM + geometry

A transparent, weighted rule-based combiner (not a second learned model, by design, so behavior stays debuggable). The weighted average of `lstm_mandown_prob` / `feature_confidence` / `persistence_score` can be overridden upward by several independent rules, each closing a specific real gap found in production testing:

1. A single very strong signal (LSTM *or* geometry) is sufficient on its own — recall stays high without requiring both sources to agree.
2. **Geometric-alone confirm**: strong, persistent geometric evidence can fully confirm MANDOWN with zero LSTM help — this is what lets an already-lying person with a cold/wrong LSTM still be caught.
3. **LSTM-alone safeguard** (mirror-image of #2): a confident LSTM, sustained over many frames, cannot confirm MANDOWN with literally no geometric grounding at all.
4. **Moderate-down-score safeguard**: a confident-but-wrong LSTM cannot ride a merely moderate (bend/sit-level) `down_score` all the way to MANDOWN.
5. **Alternate grounding path** (2D): for a foreshortened fall, `floor_proximity_score` + `height_collapse_cue` + `pose_quality` can substitute for a high `down_score` — *only* when `torso_angle` is confirmed near-zero (i.e., genuinely blind, not just unevidenced).
6. **3D grounding path** NEW this round: a confident `torso_angle_3d` reading of genuine lying, corroborated by `height_collapse_cue`, can also substitute — see §5.8-5.9.
7. **Direct down_score confirm** NEW this round: when `down_score` alone is already very strong and persistence is sustained, the fusion score is guaranteed to reach the confirm threshold directly, rather than only indirectly through a blend that can dilute it below the threshold on individual frames — see §5.10.

3.5 TrackStateMachine — temporal hysteresis

Per track_id: `NORMAL → POSSIBLE_MANDOWN` after a few consecutive "up" frames, then `→ MANDOWN` after several more consecutive "up" frames (POSSIBLE_MANDOWN and MANDOWN count identically

toward this streak). Recovery to `NORMAL` requires many consecutive genuinely-normal frames — a single flicker never clears a real alert. A short detection gap does not reset progress; a long one does.

4. The Vertical/Foreshortened Fall Problem — Why It's Hard

Every angle-based feature above (`torso_angle`, `hip_angle`, `body_axis_angle`) is computed from the 2D projection of keypoints onto the image plane. When a person's body axis points toward or away from the camera — lying with their head or feet pointed roughly at the lens — their shoulders and hips project to nearly the *same* (x, y) location, exactly as they would for someone standing upright and facing/facing-away-from the camera. There is no lateral spread to measure in either case. This is not a bug in how the angle is computed (the formula itself, `atan2(|dx|, |dy|)`, is mathematically correct for any 2D rotation) — it is a genuine information loss inherent to a single 2D projection. No amount of retuning a 2D-only formula can fix it, because the information needed (depth) was never captured by that formula's inputs in the first place.

> **The practical consequence, observed on real footage this round:** a person lying flat on the ground with their body pointed at the camera showed `torso_angle ≈ 0.02–0.22` — a reading indistinguishable from a standing person — while every other geometric signal (`down_score`, `height_collapse_cue`) was left to do all the work, and in several real postures, still weren't quite enough on their own under the pipeline's established, deliberately conservative thresholds.

5. Full Fix History (Chronological)

Each entry names the real trigger (never a hypothetical), the root cause, the fix, and its validated outcome. Reverted attempts are kept in the history rather than erased, matching this project's documentation discipline — knowing what was tried and why it failed is as valuable as knowing what worked.

5.1 V4.2 — Height-collapse cue introduced Kept

Added `height_collapse_cue`: is the observed bounding box shorter than expected for a standing person here? Driven purely by `body_height_norm`, never by any angle — the first orientation-independent signal in the system. Exposed as a corroboration source for `down_score` / `floor_proximity_score` and as an alternate grounding path in fusion.

5.2 V4.3 — Floor-region-based hip check Reverted

Idea: gate `height_collapse_cue` using a scene-wide `FloorRegion` (an aggregate band of where ankles are typically observed across the whole room), evaluated at the hip position, to exclude a standing bend from wrongly triggering the height-collapse-based alternate grounding path.

Why it was reverted: found, via real-video regression testing, to be too imprecise for real camera perspective — a genuine fall's hip does not reliably land inside an aggregate band that was never designed to be perspective-correct for every image column. This caused **two separate real regressions**: it blocked a real validation video's true positive when applied too broadly, and even after narrowing its scope, it still blocked a newly-reported real vertical-fall case.

5.3 V4.4 — Per-track, self-relative hip-drop signal Kept

Fix: replaced the scene-wide floor-region check with a signal that needs no scene calibration at all: has *this specific track's own hip* moved down substantially from where it was the last time *this same track* was confidently standing? A track with no prior standing observation (including every "already lying when first seen" case) is never gated by this — it simply isn't computed (`None`), by design, preserving the guarantee that a person found already down needs no fall history to be detected.

A sub-bug found and fixed within this same fix: using `torso_angle < threshold` alone to decide "is this track confidently standing right now" was not sufficient, because a foreshortened fall *also* reads `torso_angle ≈ 0` — meaning a track could record its own fallen posture as its "standing" baseline. Fixed by also requiring `body_height_norm` to look genuinely standing-height before trusting a frame as a baseline.

5.4 V4.5 — Fusion alternate-path torso-angle gate Kept

Real trigger: a person bending forward on their feet to reach the ground was confirmed MANDOWN on a live camera.

Root cause: the "alternate grounding path" (§3.4, rule 5) exists specifically because `torso_angle` is blind for a foreshortened fall — but it never actually verified that `torso_angle` *was* near-zero. It fired just as easily for a deep bend, where `torso_angle` is emphatically not blind (reads ~0.64) and already carries clear "not down" evidence that `down_score` / `kneeling_penalty` were supposed to apply.

Fix: require `torso_angle` to be confirmed near-zero (evidence present *and* below a small ceiling) before this path can fire at all.

5.5 V4.6 — Kneeling-penalty torso-blind fix Kept

Real trigger: a person fallen toward the camera with legs not confidently visible (ankle confidence ~0.05–0.09) was never flagged.

Root cause: `kneeling_penalty` 's occlusion fallback (used when legs aren't confidently visible) also treats a near-zero `torso_angle` as "confidently upright" — the identical blind spot as §5.4, in a different component. Measured on the real miss: this was cutting `down_score` from ~0.70 down to 0.50.

Fix: corroborate with `height_collapse_cue` the same way the fusion path does — only relent the penalty when torso is confirmed near-zero *and* independent height evidence strongly confirms a genuine collapse.

5.6 V4.7 — Combined-moderate-evidence fusion path Reverted

Real trigger: a person lying flat, legs extended, facing the camera at a shallower angle than the near-zero case above (`torso_angle=0.22`) — too shallow for the existing alternate path, but with `down_score` , `floor_proximity_score` , and `height_collapse_cue` all simultaneously moderately elevated.

Idea: let several independently-moderate, orientation-immune signals combine into confirmation when sustained.

> **Why it was reverted:** a synthetic crouching posture (no real bend, just kneeling penalty near zero) measured comparable or higher numbers on the same three signals, and reached MANDOWN under a confident LSTM once this path was added — verified before shipping via the project's own negative-control testing discipline. No threshold on these three signals alone could admit the real target case without also reopening this crouching false positive.

5.7 V4.8 — Monocular 2D→3D pose lifting Kept — the structural fix

This is the headline addition of this round: a new module (`src/pose_3d.py`) wraps MediaPipe's Pose Landmarker to recover an approximate 3D skeleton from the same single 2D image, per detected person, per frame.

How it works

- Crops the YOLO bounding box (with padding — BlazePose expects to see a full body with margin) and runs MediaPipe's landmarker on just that crop.
- Reads `pose_world_landmarks` — a metric, hip-centered 3D estimate. Verified empirically against real footage (not just assumed from documentation) that **Y increases downward** for a normally-oriented camera view: confirmed on a real standing person (torso vector dominated by +Y) and a real, clearly-lying-flat person (angle from the Y axis ~77–86°).
- Computes `torso_angle_3d` = angle between the real 3D shoulder→hip vector and vertical, normalized to the same 0–1 convention as the existing 2D `torso_angle`.

Why this is not "just another 2D feature"

MediaPipe's estimate is an *inference* from a learned prior over plausible human body configurations (trained on motion-capture datasets), not a physical depth measurement — it does not always agree with 2D features, and its accuracy on an unusual camera angle is not guaranteed by documentation alone. This is why every claim about it in this report is backed by a real, measured example, not an assumption.

Performance gate

Because this is a second model inference per person per frame, it does not run unconditionally. It only runs when the cheap 2D height signal already looks suspicious (`body_height_norm < 0.85`) or is not yet calibrated at all — never on every normal standing frame in a live deployment.

Where it plugs in

Two places, both requiring a real geometric floor alongside the 3D reading (a lone 3D reading can never confirm by itself):

- `kneeling_penalty` : a confident `torso_angle_3d` reading of lying can relent the penalty directly, independent of the 2D `torso_angle` reading.
- `fusion.py` 's new "confident 3D lying" path: confirms MANDOWN together with minimal geometric backing, independent of the 2D-derived alternate path.

5.8 V4.8.1 — Height-collapse requirement added to the 3D path Kept

Real trigger: the exact same deep-bend posture from §5.4/§5.5 reached MANDOWN again through the *new* 3D path, because a sufficiently deep bend genuinely tilts the torso in real 3D space too — measured live at `torso_angle_3d=0.777` , almost identical to its own 2D `torso_angle=0.778` . `torso_angle_3d` is not blind to a bend the way it is to a foreshortened fall, so a high reading alone does not distinguish the two.

Fix: require `height_collapse_cue` to also confirm a genuine collapse before trusting the 3D path — a bend can never bring the hip down toward the floor (legs stay extended, hip stays at standing height), whereas any genuine fall does.

5.9 V4.9 — Torso-hip-angle-mismatch signal Attempted twice; kept, with caveats

Real trigger: even the height-collapse requirement above was not enough — under real per-camera calibration, a deep bend's overall bounding-box height *also* genuinely shrinks (head and shoulders come down close to hip level), so `height_collapse_cue` cleared its own floor too (measured: 0.53).

The real discriminator found: in a bend, the legs stay planted and near-vertical (low `hip_angle`) while only the torso hinges over (high `torso_angle`) — a large gap between the two. Genuine diagonal lying tilts the whole body as one unit, so `torso_angle` and `hip_angle` read close together (verified: ~0.02 gap on a diagonal-lying test fixture, vs. ~0.55 on the real bend).

> **First attempt — a scare, later resolved:** checked against this project's single most protected real true-positive track using a *test script that did not wire in real per-camera height calibration*. That flawed check appeared to show the entire confirmed event suppressed to zero. The change was reverted immediately out of an abundance of caution. Re-verifying with the camera's *real* saved calibration wired in showed the actual confirming frames of that track have a *small* torso-hip gap (0.01–0.23) — the uncalibrated test had been misleading. The fix was re-applied after this correction, and the project's full test suite (250 tests) plus this targeted recheck both passed.

This episode is recorded in detail because it illustrates the project's validation discipline in action: a change was reverted the moment a check — even an imperfect one — suggested regression risk to the most safety-critical case, and only re-applied after the check itself was corrected and re-verified.

5.10 V4.10 — Direct down_score confirm path Kept — closed 4 gaps at once

Real trigger: a person lay flat, foreshortened toward the camera, for a sustained 42 real seconds. The event log showed the fusion engine's own per-frame evidence read `suggested_tier: "MANDOWN"` with `down_score=0.925` (near-maximal) — yet the event never confirmed. The LSTM was 99.9% confident this was "normal" (`lstm_mandown_prob=0.001`) — the same 2D blind spot affecting the LSTM's own 5-feature input, since it is computed the same way.

Root cause: the existing "geometric-alone confirm" override (§3.4, rule 2) requires `feature_confidence ≥ 0.60` — but `feature_confidence` is a *blend* of `down_score` with several other components (including `fall_motion_score`, which correctly reads ~0 for a static lying person) and is then discounted by `pose_quality`. In this real case the blend landed at 0.602 — a hair above the threshold, not comfortably above it. Any small frame-to-frame dip broke the override, preventing the state machine's required run of consecutive confirming frames.

Fix: give `down_score` alone — already the single most direct "is this person down" signal, and already sufficient on its own to bypass the LSTM safety cap — a *direct* path to force the fusion score to the confirm threshold, instead of only reaching it indirectly through a blend that can dilute it.

> **This is a monotonic improvement:** it can only *raise* the fusion score for frames where `down_score` and persistence were already independently strong — it cannot lower the score for

> any case, and cannot admit any case that wasn't already permitted to bypass the safety cap by the pre-existing rule. As a side effect, it also closed three other long-standing, honestly-documented "known limitation" gaps in the test suite (diagonal lying, side-lying with bent knees, and the calibrated vertical fall with a cold LSTM) — all four shared the exact same root cause.

5.11 Ground-plane 2D perspective calibration upgrade Kept

Independent of the fall-orientation fixes above: the scene calibration's perspective fit was upgraded from a 1-D (image row only) linear regression to a proper 2-D (row and column) fit. A camera mounted at an angle has perspective scale that genuinely varies across both axes, not just vertically — most visible on a wide-angle lens, where the old row-only fit's "expected size" was only correct along one column and increasingly wrong toward the frame edges. Degrades gracefully to the old row-only behavior when there isn't yet enough horizontal variation in the observed samples to fit a stable column term.

5.12 Other issues found and fixed this round (unrelated to orientation, listed for completeness)

Issue	Fix	Status
Red MANDOWN box "stuck on screen" after the person is gone	Persistence window made configurable, reduced from a hard-coded 10s to 3s	Fixed
Multiple simultaneously-fallen people only got one red box in the saved screenshot	Screenshot save deferred until every person in the frame is classified, then draws everyone currently "fallen"	Fixed
A stationary object (a dark bean-bag cushion) misdetected as a person, then classified as falling	Root-caused to the YOLO pose detector itself, not the MANDOWN scoring logic; see §7	Open — see §7

6. Real-World Validation Results

6.1 The four standing reference videos

Video	Confirmed	Confidence	Notes
`mandown_2.mp4`	1 / 1	**`0.881`**	Boosted from ~0.62 pre-3D-lifting
`man_down_c08.mp4`	1 / 1	**`0.841`**	Boosted from ~0.65
`man_down_zoomed_demo.mp4`	1 / 1	**`0.867`**	9 different real people tracked across the video; zero false positives among the rest
`M_D_2.mp4` — protected track	1 / 1	**`0.831`** / $\text{down}_{\text{score}}$**`=0.813`**	**Exact match** to the long-established protected baseline for this track — zero regression across every change in this round

6.2 Live RTSP testing — postures confirmed working

- A person lying flat, foreshortened toward the camera, arms folded — confirmed via the 3D grounding path (`torso_angle_3d=0.575` , `height_collapse_cue=0.63`).
- The same general posture sustained for 42 real seconds — confirmed via the V4.10 direct `down_score` path after the fix (`down_score=0.925` measured in the real event log).
- A person lying laterally (side-on to the camera) — confirmed cleanly via the pre-existing 2D path, unaffected by any change this round.

6.3 Live RTSP testing — false positives found and fixed

- A person bending forward on their feet to reach a low table/device (§5.4, §5.5, §5.8, §5.9) — now correctly stays below MANDOWN across a confident LSTM sweep, verified directly against the real captured keypoints on repeat occurrences of the same posture.
- The closest, most foreshortened real standing person available in the entire test corpus (a worker in the "zoomed" close-camera reference video) — directly measured at `torso_angle_3d=0.243` , correctly read as upright, staying at NORMAL/POSSIBLE even under a maximally confident-wrong LSTM (0.95).

7. Known Remaining Limitations (Honest Accounting)

Per this project's standing discipline: a gap is reported honestly with real numbers, never quietly patched by loosening an unrelated threshold.

7.1 OPEN — Lying on the back with one knee raised

Most recently found. A person lay foreshortened toward the camera with one knee bent upward (hands clasped on chest) — the geometry differs enough from the flat-arms-folded posture in §6.2 that it did not confirm. Measured directly on the real frame:

```
torso_angle       = 0.121   (2D — near-zero, foreshortening blind spot as expected)
hip_angle         = 0.175
torso_angle_3d    = 0.319   (below the 0.55 "confident lying" floor — the raised
knee likely
                             shifts MediaPipe's inferred 3D shoulder/hip positions)
down_score        = 0.454   (below the 0.65 direct-confirm floor)
floor_proximity   = 0.552   (below the 0.70 floor the 2D alternate path needs)
height_collapse   = 0.649   (clears its own floor, but no path currently uses it
alone here)
→ fusion capped at 0.45 (POSSIBLE_MANDOWN ceiling) regardless of LSTM confidence
```

Diagnosis: none of the three "escape valves" (2D alternate path, 3D grounding path, direct down_score path) individually clears its own bar for this specific posture, even though every individual number is elevated in the correct direction. This is structurally the same class of problem as the V4.7 attempt (§5.6) — several moderately-elevated signals, no single one strong enough — and should **not** be patched by simply lowering one of the existing floors, for the same reason V4.7 was reverted: the same combination of moderate readings is also producible by postures that should not confirm. A safe fix needs a new, targeted discriminator specific to this posture (most likely: understanding why `torso_angle_3d` reads so much lower here than in the flat-arms case despite both being genuine lying-toward-camera postures — possibly the raised knee's keypoints are pulling MediaPipe's inferred hip position, or a per-posture MediaPipe accuracy limitation), analogous to how the hip-angle-mismatch signal (§5.9) was found for the bend case.

7.2 OPEN — no safe algorithmic fix identified — Inanimate object misdetected as a person

A dark bean-bag cushion on a couch is detected by the YOLO pose model as a "person" with moderate confidence (~0.63), and — once treated as a person — is reasonably (if wrongly) read as a collapsed posture by the downstream logic.

Why raising the detection-confidence threshold does not work: real confirmed falls in the reference videos are detected with confidence as low as 0.30–0.35 during parts of their own event (a collapsed, occluded, or unusually-posed real person is often harder for YOLO to confidently detect than this static object is). Any threshold high enough to reject the object would also reject real falls.

Why cross-checking with the new 3D lifting does not help either: tested directly — MediaPipe also confidently hallucinates a lying pose on the same object (`torso_angle_3d=0.89` , `evidence_quality=0.995`). Both models are pattern-matching plausible human shapes, not verifying "is this actually a human", so this cross-check adds no new information for this specific failure mode.

Recommended fix: physical — move, remove, or reposition the object (or reangle the camera slightly) so it no longer presents a person-like silhouette. No downstream scoring change can reliably separate "a real, oddly-posed person" from "an object that happens to look like one" using only the confidence signal available, since the two populations genuinely overlap.

7.3 OPEN — Camera 2, shallow-angle lying case

Found earlier in this engineering effort, prior to the 3D-lifting work: a person lying at a shallower angle than the near-zero foreshortened case, where every individual 2D/height signal reads moderate rather than strong. Documented at the time with exact numbers; not yet re-tested against the current, fuller feature set (2D+3D combined) — worth revisiting now that more discriminators exist.

7.4 Environment, not code — GPU clock throttling observed

During validation, the GPU was observed stuck at 180 MHz of a 3090 MHz maximum despite showing 100% utilization — an infrastructure/power-management issue unrelated to any change in this report, but worth investigating separately if real-time throughput becomes a concern in deployment, since it affects every video/live-stream run equally.

8. Configuration Reference

All thresholds discussed in this report live in `configs/mandown_config.yaml` (loaded into `src/config.py` dataclasses). Every value carries an in-file comment explaining its origin — real measured number, synthetic fixture, or design margin — so a future engineer can tell at a glance whether a given number is safe to retune or load-bearing for a specific protected case.

Config section	Governs
`hip_drop`	The per-track self-relative hip-drop signal (§5.3)
`kneeling`	`kneeling_penalty` and all its corroboration/relent conditions (§5.5, §5.8, §5.9)
`pose_3d`	Monocular 2D→3D lifting: model path, confidence floors, the real-time performance gate (§5.7)
`fusion`	All EvidenceFusion override/cap rules (§3.4)
`pose_failure.occluded_marker_seconds`	How long a lost track's red box lingers on screen (§5.12)

9. Files Changed This Round

File	Change
`src/pose_3d.py`	**New.** Monocular 2D→3D pose lifting via MediaPipe.
`models/mediapipe/pose_landmarker_lite.task`	**New.** Downloaded MediaPipe model asset.
`src/scene_profile.py`	Perspective fit upgraded from 1-D (row) to 2-D (row + column).
`src/feature_extraction.py`	New `torso_angle_3d` / `torso_angle_3d_evidence_quality` fields; `ankle_x` and `pose_3d` threaded through `extract()`.
`src/feature_mandown.py`	`_kneeling_penalty` : 3D-lifting relent path, torso-hip-angle-mismatch signal.
`src/fusion.py`	3D grounding path; direct `down_score` confirm path; torso-hip-mismatch exclusion.
`src/pipeline.py`	Wires `Pose3DEstimator` into the per-person loop with the performance gate; ankle_x computed before feature extraction.
`src/config.py` , `configs/mandown_config.yaml`	New `Pose3DConfig` ; new thresholds for every fix above, each with a rationale comment.
`requirements.txt`	Added `mediapipe` .
`tests/test_pose_3d.py`	**New.** Unit + real-model integration tests for the lifting module.
`tests/test_fusion.py` , `tests/test_feature_mandown.py` , `tests/test_already_lying_integration.py` , `tests/test_scene_profile.py`	Extensive new/updated regression coverage — every real number in this report has a corresponding unit test.

Full test suite as of this report: **250 passed, 0 failed, 0 unexpectedly skipped.**

10. Recommendations / Next Steps

1. **Highest priority:** investigate §7.1 (raised-knee lying posture) with the same rigor as the bend case — capture more real examples of this posture, measure exactly why `torso_angle_3d` reads low, and design a targeted fix validated against both this case and the existing protected true positives before shipping.
2. Re-test §7.3 (Camera 2 shallow-angle case) against the current, fuller feature set — the 3D-lifting work may have already narrowed or closed this gap as a side effect, the way V4.10 did for three other cases.
3. For §7.2 (object misdetection), the recommended path is physical (reposition the object/camera) rather than further algorithmic tuning, given the demonstrated overlap between real low-confidence falls and this object's detection confidence.
4. Investigate the GPU clock-throttling observation (§7.4) separately if live-deployment frame rate becomes a concern.
5. Continue the established validation discipline for any future change: unit tests with real measured numbers, explicit negative-control checks against every previously-fixed false positive, and re-verification of the protected `M_D_2.mp4` track before considering any fusion/scoring change complete.

MANDOWN V5 Implementation Report — generated from a live engineering session's real RTSP camera testing, unit test suite, and reference-video validation. All numeric values quoted in this report were measured directly against real footage or the project's own automated test suite at the time of writing, not estimated.